

# Extension of the RSA Production Model

## Mechanism-level exploration and the $2 \times 2$ tradeoff

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2026-04-15

Branch feat/extension-v5 • dc subset ( $N = 3196$ ) • 16 commits

# Where we start: the paper-reported model

Incremental recursive speaker, hierarchical per-participant  $\alpha$  offsets. **Theory-driven**: minimal parameters with clear motivation.

## The $2 \times 2$ story (memo §1)

| Model                 | ELPD LOO | $\Delta$ ELPD | dSE   | $\Delta$ /dSE | $p$    |
|-----------------------|----------|---------------|-------|---------------|--------|
| <b>Inc. recursive</b> | -7180    | 0             | —     | —             | —      |
| Inc. static           | -7199    | -19           | 1.84  | -10.28        | < .001 |
| Global static         | -7626    | -446          | 40.27 | -11.08        | < .001 |
| Global recursive      | -7628    | -448          | 40.69 | -11.02        | < .001 |

- **Speaker architecture dominates**: inc  $\gg$  glb ( $\Delta$ /dSE  $\approx 11$ ).
- **Semantic regime within incremental**: recursive  $>$  static ( $\Delta$ /dSE = 10.28).
- **Semantic regime within global**: null.

## But: the fit is poor

- $R^2 = 0.346$  across the 90 (condition  $\times$  utterance) cells.
- **L1 aggregate residual = 3.17** (cells with emp  $\geq 2\%$ ).
- Lapse  $\varepsilon = 0.22$  — nearly a quarter of responses as uniform noise.

### Seven systematic misfits (memo §5.3)

| Condition         | Sharpness | Utterance | Emp  | Model | $\Delta$     |
|-------------------|-----------|-----------|------|-------|--------------|
| colour-sufficient | high      | C         | .716 | .403  | <b>-.313</b> |
| size-sufficient   | low       | SF        | .365 | .081  | <b>-.284</b> |
| colour-sufficient | low       | C         | .628 | .402  | <b>-.225</b> |
| both-necessary    | low       | C         | .009 | .222  | <b>+.214</b> |
| size-sufficient   | high      | SFC       | .022 | .211  | <b>+.189</b> |
| both-necessary    | high      | C         | .000 | .175  | <b>+.175</b> |
| size-sufficient   | low       | SFC       | .032 | .187  | <b>+.155</b> |

# Diagnosis (memo §5.3)

Residuals decompose along two independent axes.

## 1. Context-dependence of 1-word C

- Under-predicted by  $\sim .25$ – $.31$  in colour-sufficient
- Over-predicted by  $\sim .18$ – $.21$  in both-necessary
- No single  $\alpha_C$  can fix both.

## 2. Length residuals: saturation, not level shift

- 2-word (SF) under-predicted by  $\sim .28$  in blurred size-sufficient
- 3-word (SFC) over-predicted by  $\sim .17$  in size-sufficient
- No monotone change to a linear  $\gamma$  fixes both.

**These diagnostics motivate the mechanism track.**

# Extension timeline (dc subset, $N = 3196$ )

erdc, zrdc, brdc — size  $\times$  colour distinguishing, form redundant.

Three relevant\_property values inside: **first** (size sufficient), **both**, **second** (colour sufficient).

| Step           | Mechanism                             | New parameters                                      | Literature cue                  |
|----------------|---------------------------------------|-----------------------------------------------------|---------------------------------|
| ext-v1         | per-dim $\alpha, \gamma, \varepsilon$ | $\alpha_D, \alpha_C, \alpha_F, \gamma, \varepsilon$ | dim-salience asymmetry          |
| <b>v5a</b>     | condition-gated colour boost          | $\lambda_C$ (step-0 only)                           | Pechmann 89; Rubio-Fernández 16 |
| <b>v5b</b>     | saturating length bias                | $\gamma_1, \gamma_2$                                | overspec not monotone           |
| <b>v5 (F1)</b> | + csuff length offsets                | $\delta\gamma_1, \delta\gamma_2$                    | context-dep. length bias        |
| <b>v5 (C1)</b> | + canonical-order penalty             | $\mu_{nc}$                                          | Sproat 91; Cinque 94; Scott 02  |
| <b>v5 (F2)</b> | + sharpness-dependent length          | $\eta_1, \eta_2$                                    | overspec > under blur           |

Flag correction along the way: COLOUR\_SUFFICIENT\_CONDITIONS was first (*ercf*, *zrcf*, *brcf*); corrected to (*ercf*, *zrdc*) — only trials where colour alone identifies the target.

## Stage 1: ext-v1 (already in memo §4.1)

Per-dimension  $\alpha$  + explicit  $\gamma$  + lapse  $\varepsilon$ .

|               | ELPD  | vs baseline |               | $R^2$ | L1   |
|---------------|-------|-------------|---------------|-------|------|
| baseline      | -7157 | 0           | baseline      | .35   | 3.17 |
| <b>ext-v1</b> | -6387 | +770        | <b>ext-v1</b> | .60   | 3.17 |

Posterior:  $\alpha_D = 6.94$ ,  $\alpha_C = 1.84$ ,  $\alpha_F = 1.88$ ,  $\gamma = 2.32$ ,  $\varepsilon = 0.22$ .

**But:** seven memo §5.3 misfits persist;  $\varepsilon = 0.22$  uncomfortably high.

## Stage 2a: $\lambda_C$ alone (v5a)

Step-0 logit boost for **C** when `is_colour_sufficient = 1`.

|            | ELPD  | $\Delta$ vs ext-v1 | $\lambda_C$ posterior |
|------------|-------|--------------------|-----------------------|
| ext-v1     | -6387 | 0                  | —                     |
| <b>v5a</b> | -6265 | +122               | 3.4 (95% CI 2.8–4.0)  |

- Strong positive  $\lambda_C$ : colour gets extra first-step pull when it suffices.
- second/sharp C: emp .716, was .403, now  $\sim$  .55.
- **CF still too high** in second: boost spreads to C-initial compounds.

## Stage 2b: saturating $\gamma$ alone (v5b)

Replace linear  $\gamma \cdot k$  with  $\gamma_1 \mathbb{I}[k \geq 1] + \gamma_2 \mathbb{I}[k \geq 2]$ .

|        | ELPD  | $\Delta$ vs ext-v1 |
|--------|-------|--------------------|
| ext-v1 | -6387 | 0                  |
| v5b    | -6381 | +6                 |

**Essentially no improvement without  $\lambda_C$ .**

The two mechanisms are **synergistic**:  $\gamma$ -saturation only matters after C-initial mass is correctly apportioned.



## Stage 3: F1 — condition-dependent length ( $v5 + \delta\gamma$ )

Let the length bias **shrink** on colour-sufficient trials:

$$\gamma_{1,\text{eff}} = \gamma_1 + \delta\gamma_1 \cdot \text{is\_csuff}.$$

|                | ELPD  | $\Delta$ vs ext-v1 |
|----------------|-------|--------------------|
| ext-v1         | -6387 | 0                  |
| v5a            | -6265 | +122               |
| <b>v5 (F1)</b> | -5933 | +454               |

Posteriors:  $\lambda_C = 5.0$ ,  $\gamma_1 = 2.56$ ,  $\gamma_2 = 0.92$ ,  $\delta\gamma_1 = -1.89$ ,  $\delta\gamma_2 = -3.51$ ,  $\epsilon = 0.16$ .

**Effective  $\gamma$  on csuff trials:** 0.67,  $-2.59$  — 3-word heavily penalised, short utterances favoured.

$R^2 = .67$ ,  $L1 = 2.70$ . **Remaining puzzle:** over-predicts DC and DFC, under-predicts DCF.

## Stage 4: C1 — canonical-ordering penalty

Add  $\mu_{\text{nc}} \cdot \mathbb{I}[\text{utterance violates } D < C < F]$ .

Non-canonical set: CD, CDF, CFD, DFC, FD, FDC, FC, FCD.

|                | ELPD         | $\Delta$ vs ext-v1 |
|----------------|--------------|--------------------|
| v5 (F1)        | -5933        | +454               |
| <b>v5 (C1)</b> | <b>-5323</b> | <b>+1063</b>       |

$\mu_{\text{nc}} = -5.07$  — very strong canonical-English preference.

$\varepsilon$  drops to **0.10**.

$R^2 = .91$ ,  $L1 = 1.34$ .

7 of the 8 user-flagged residuals fixed: DCF, DF, DC, DFC, C.

**Side-effect:** on first/sharp, bare D under-predicted (.082 vs emp .269) and DF over-predicted.

## Stage 5: F2 — sharpness-dependent length (final)

Add  $\eta_1, \eta_2$  offsets active on `is_sharp` trials.

|                | ELPD  | $\Delta$ vs ext-v1 | $R^2$ |
|----------------|-------|--------------------|-------|
| v5 (C1)        | -5323 | +1063              | .91   |
| <b>v5 (F2)</b> | -5294 | +1093              | .94   |

Posteriors:  $\eta_1 = -0.61$ ,  $\eta_2 = -0.34$  (both informatively negative).

Effective  $\gamma$  on **sharp**: 1.71, 1.24 vs on **blurred**: 2.32, 1.58.

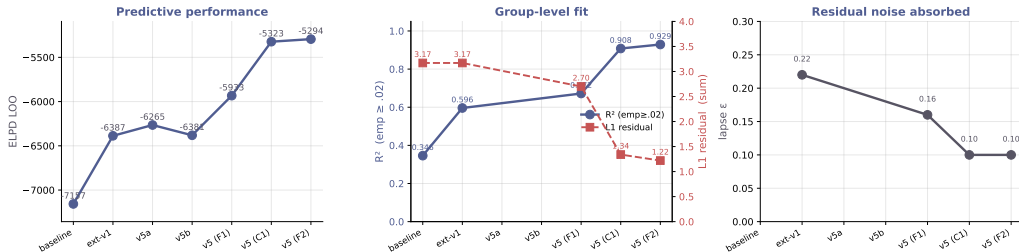
$\varepsilon$  stays at 0.10; 0 divergences; all  $\hat{R} \leq 1.02$ .

**Stubborn residual**: first/sharp D still under-predicted (.269  $\rightarrow$  .102).

Likely strategy heterogeneity: some participants minimum-describe, others hedge.

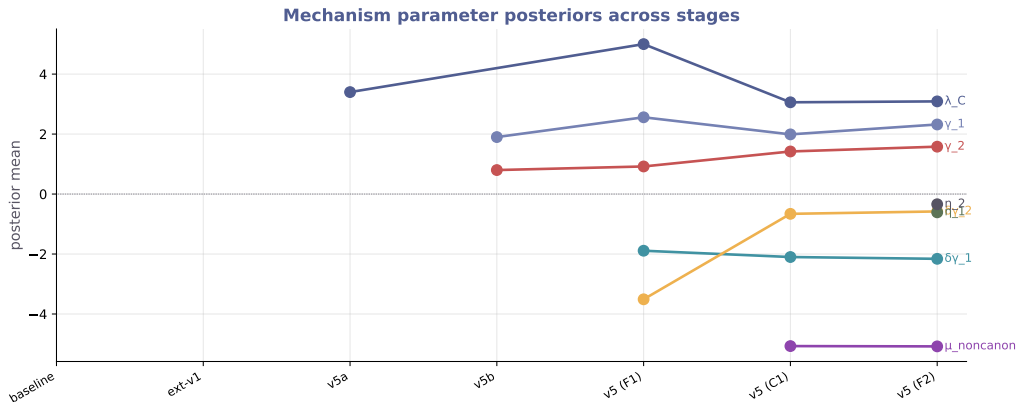
# Fit evolution across stages

Extension v5 — Fit evolution across mechanism additions (dc subset, N = 3196)



Each new mechanism shifts LOO and  $R^2$  upward and  $\epsilon$  downward. v5b (sat  $\gamma$  alone) flat vs ext-v1.

# Mechanism posteriors across stages



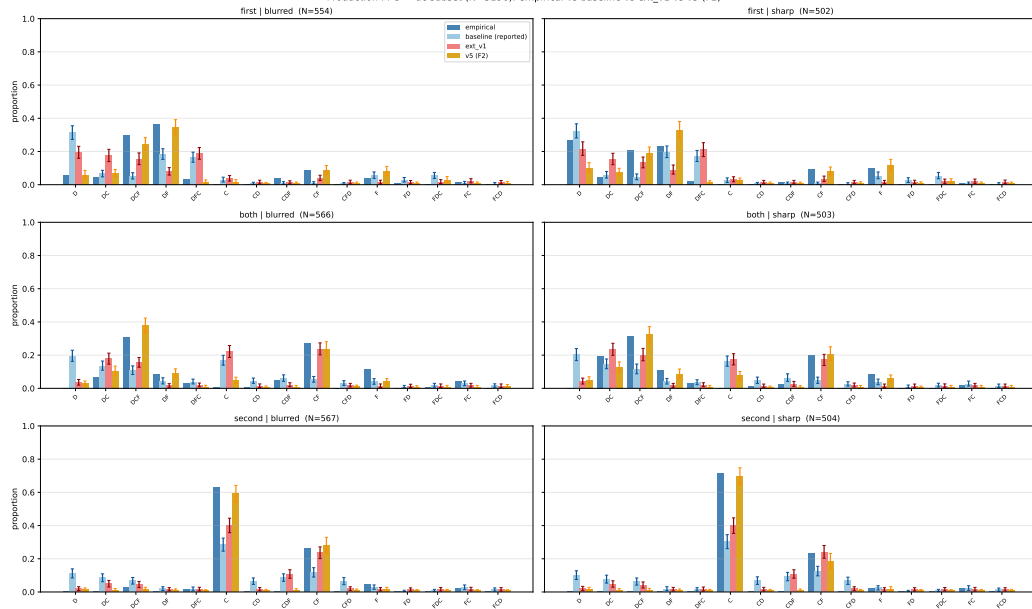
Each parameter enters with a clear posterior sign and settles by F2.  $\lambda_C$ ,  $\mu_{\text{nc}}$  strong;  $\eta$  modest;  $\delta\gamma_1$  strongly negative on csuff;  $\delta\gamma_2$  shrinks once  $\mu_{\text{nc}}$  absorbs canonical-order variance.

## Final v5 (F2) parameters — full record

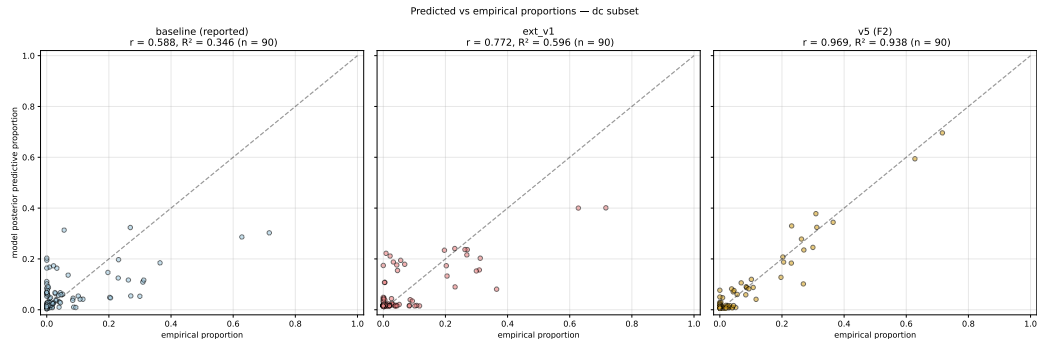
| Parameter        | Prior      | Posterior mean (95% CI) | $\hat{R}$   |
|------------------|------------|-------------------------|-------------|
| $\alpha_D$       | HalfN(5)   | (hier. w/ $\delta_p$ )  | $\leq 1.01$ |
| $\alpha_C$       | HalfN(5)   | (hier.)                 | $\leq 1.01$ |
| $\alpha_F$       | HalfN(5)   | (hier.)                 | $\leq 1.01$ |
| $\lambda_C$      | N(0,1)     | +3.09 (2.67, 3.55)      | 1.00        |
| $\gamma_1$       | N(0,1)     | +2.32 (2.16, 2.49)      | 1.00        |
| $\gamma_2$       | N(0,1)     | +1.58 (1.43, 1.74)      | 1.01        |
| $\delta\gamma_1$ | N(0,1)     | -2.16 (-2.33, -1.97)    | 1.00        |
| $\delta\gamma_2$ | N(0,1)     | -0.58 (-1.21, +0.04)    | 1.00        |
| $\eta_1$         | N(0,1)     | -0.61 (-0.77, -0.43)    | 1.00        |
| $\eta_2$         | N(0,1)     | -0.34 (-0.51, -0.15)    | 1.00        |
| $\mu_{nc}$       | N(0,1)     | -5.08 (-5.61, -4.56)    | 1.01        |
| $\log \beta$     | N(0,0.5)   | -0.17                   | 1.00        |
| $\varepsilon$    | Beta(1,50) | 0.10 (0.09, 0.12)       | 1.00        |
| $\tau$           | HalfN(0.2) | 0.62 (0.51, 0.73)       | 1.00        |

# PPC barplot — baseline vs ext-v1 vs v5 (F2)

Production PPC — dc subset (N=3196): empirical vs baseline vs ext\_v1 vs v5 (F2)



# Predicted vs empirical ( $n = 90$ cells)



$R^2$ : baseline .35  $\rightarrow$  ext-v1 .60  $\rightarrow$  **v5 (F2) .94.**



# The $2 \times 2$ question: does the paper story survive?

Under F2 mechanisms we re-ran the  $2 \times 2$  (speaker architecture  $\times$  semantics regime) on the dc subset.

Three parameterisations to compare:

- 1 **Paper-reported (memo §1)**: minimal, theory-driven.
- 2 **F2-fixed globals**: global cells use  $\gamma/\delta\gamma/\eta/\mu_{nc}$  fixed at inc-recursive posterior means.
- 3 **F2-full globals**: every parameter sampled freely in every cell. Fairest comparison.

## 2×2: paper-reported

|                                   | recursive | static | $\Delta(\text{rec} - \text{sta})$                |
|-----------------------------------|-----------|--------|--------------------------------------------------|
| incremental                       | -7180     | -7199  | +19 ( $\Delta/\text{dSE} = 10.28$ )<br>-2 (null) |
| global                            | -7628     | -7626  |                                                  |
| $\Delta(\text{inc} - \text{glb})$ | +448      | +427   |                                                  |
| $\Delta/\text{dSE}$               | 11.02     | 11.08  |                                                  |

### Paper narrative

- Architecture dominates:  $\text{inc} \gg \text{glb}$ ,  $\Delta/\text{dSE} \approx 11$ .
- Recursive helps incremental a lot, does not help global.
- Diff-in-diff = +21 (inc benefits more from rec).

## 2×2: F2 with globals' params fixed at inc posterior means

|                                 | recursive | static | $\Delta(\text{rec}-\text{sta})$                |
|---------------------------------|-----------|--------|------------------------------------------------|
| incremental                     | -5294     | -5297  | +2.80 ( $\Delta/\text{dSE} = 2.22$ , marginal) |
| global (fixed)                  | -5600     | -5619  | +19.15 ( $\Delta/\text{dSE} = 6.99$ )          |
| $\Delta(\text{inc}-\text{glb})$ | +306      | +323   |                                                |
| $\Delta/\text{dSE}$             | 10.51     | 11.03  |                                                |

- Architecture still dominant ( $\Delta/\text{dSE} \approx 10.5$ ).
- **Interaction has flipped**: rec now helps global, not incremental.
- But global's length/order parameters are fixed at inc values — unfair test.

## 2×2: F2 with globals fully sampled — the fair picture

|                          | recursive | static | $\Delta(\text{rec-sta})$             |
|--------------------------|-----------|--------|--------------------------------------|
| incremental              | −5294     | −5297  | +2.80 ( $\Delta/\text{dSE} = 2.22$ ) |
| global (full)            | −5357     | −5365  | +8.25 ( $\Delta/\text{dSE} = 3.82$ ) |
| $\Delta(\text{inc-glb})$ | +63       | +69    |                                      |
| $\Delta/\text{dSE}$      | 3.18      | 3.48   |                                      |

### Key changes vs paper-reported

- Architecture effect  $\Delta/\text{dSE}$ :  $11 \rightarrow 3.2$  ( $\sim 1/3$  of paper's).
- Regime-within-inc  $\Delta/\text{dSE}$ :  $10.28 \rightarrow 2.22$  ( $\sim 1/5$  of paper's).
- Regime-within-glb:  $\approx 0 \rightarrow 3.82$  (appears, modestly).
- Diff-in-diff:  $+21 \rightarrow +5.45$  (same sign, small magnitude).

## Full record — side-by-side

|                                 | Paper       | F2-fixed          | <b>F2-full</b>  |
|---------------------------------|-------------|-------------------|-----------------|
| $R^2$ (all cells)               | .35         | .94               | .94             |
| L1 residual                     | 3.17        | 1.22              | 1.22            |
| Lapse $\varepsilon$             | 0.22        | 0.10              | 0.10            |
| Arch effect $\Delta/\text{dSE}$ | $\pm 11$    | $\pm 10.5$        | $\pm 3.2$       |
| Regime  inc $\Delta/\text{dSE}$ | 10.28       | 2.22              | 2.22            |
| Regime  glb $\Delta/\text{dSE}$ | $\approx 0$ | 6.99              | 3.82            |
| Interaction (diff-in-diff)      | +21         | -16               | +5.45           |
| Rec helps. . .                  | incremental | global (artefact) | slightly global |

# The tradeoff

## Theory track (paper)

- Minimal, motivated mechanisms
- Clear  $2 \times 2$  interaction
- **Poor group-level fit ( $R^2 = .35$ )**
- **Seven large misfits in §5.3**
- Lapse absorbs unexplained structure

## Mechanism track (F2)

- Rich; each term literature-grounded
- Excellent fit ( $R^2 = .94$ )
- All seven misfits resolved
- Lapse halved ( $0.22 \rightarrow 0.10$ )
- **Architecture signal shrinks to  $\Delta/dSE = 3$**
- **Regime-within-inc weakens to marginal**

**Core tension.** Mechanisms designed to reduce residuals explain variance that the paper's minimal parameterisation attributed to **architecture**  $\times$  **regime**. The interaction survives in reduced form.

# Two readings

## Reading I: mechanisms as rivals

The paper's interaction was partly *architectural sparsity fitting error structure*. Once error is absorbed by explicit mechanisms (colour salience, canonical ordering, context-dep. length), the residual architectural signal is modest. The main claim must weaken.

## Reading II: mechanisms as scaffolding

$\lambda_C$ ,  $\mu_{nC}$ ,  $\delta\gamma$ ,  $\eta$  are *implementations* of what context-updating  $\times$  incremental architecture is *supposed* to do. Of course adding them absorbs architectural signal — they are the phenomena the architecture was indirectly capturing.

**Both are defensible. The choice shapes the paper's framing.**

# Four concrete decisions

## 1 Report v5 (F2) or keep the paper-reported model?

A middle path: main-text keeps the reported model; supplementary section shows F2 as mechanism-level refinement.

## 2 If we adopt F2, how do we frame the interaction?

Reading I softens; Reading II preserves. A referee will ask.

## 3 Adopt the corrected COLOUR\_SUFFICIENT flag (*ercf*, *zrdc*)?

Independent of the F2 decision. Probably yes — the definition is simply more accurate.

## 4 Generalise beyond the dc subset?

Parallel runs on df (size+form) and cf (colour+form) would test whether F2 mechanisms are symmetric or colour-specific.



# Open questions

- **Strategy heterogeneity.**  $\varepsilon = 0.10$  is smaller than ext-v1's 0.22 but non-zero. The first/sharp bare-D vs DF split looks like two participant sub-populations.
- **Full-data ( $N = 9586$ ) run.** Our early full-data run used the incorrect flag. A clean full-data F2 run would take  $\sim 15$  min on GPU.
- **df and cf subsets.** Do the mechanisms generalise? If a “form-sufficient” flag is needed, symmetric structure; if not, colour-specificity.
- **LOO warnings.** `az.compare` flagged Pareto  $k > 0.7$  in some cells. Differences reliable in direction, noisier in magnitude.

# Summary

- Paper-reported model: theory-driven, interpretable  $2 \times 2$ , but  $R^2 = .35$ .
- Residuals trace to three independent phenomena: **colour salience**, **canonical ordering**, **condition-dependent length bias**.
- Adding all three (v5 F2) lifts fit to  $R^2 = .94$  and lapse to .10, with every new parameter informative, 0 divergences.
- Re-running the  $2 \times 2$  under F2 with **freely sampled** globals: architecture effect shrinks  $\sim 3\times$ ; regime effect shrinks and partially flips direction.
- Two readings of the tradeoff; both defensible.
- Decision the paper needs: **report F2, report paper-reported, or both?**

Artefacts: `branch feat/extension-v5`, spec at `docs/superpowers/specs/...`, synthesis at `10-writing/extension_v5_synthesis.md`.

# Thanks

## Questions and discussion.

feat/extension-v5

10-writing/extension\_v5\_synthesis.md

10-writing/extension\_v5\_slides\_csp.tex

05-modelling-production-data/figures/v5/

05-modelling-production-data/results/

16 commits

detailed write-up

this deck

figures used above

LOO + residual CSVs